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Guided Critique of Evaluation Article

A Vaping Cessation Text Message Program for Adolescent E-Cigarette Users: A Randomized Clinical Trial

Purpose

The purpose of the study by Amanda L. Graham and colleagues was to evaluate the effectiveness of a text-message vaping cessation intervention for adolescents who use e-cigarettes. The study specifically examined whether adolescents participating in the "This is Quitting" text-message program were more likely to stop vaping compared to adolescents in an assessment-only control group. The researchers aimed to determine whether a tailored and interactive mobile health intervention could improve vaping cessation rates among teenagers interested in quitting nicotine vaping.

The program being evaluated was "This is Quitting," an automated text-message intervention developed to support adolescents and young adults attempting to quit vaping. The program provided coping strategies, social support, motivation, and behavioral skills training through personalized text messages. Messages were tailored to participants' quit dates, vaping habits, and emotional support needs. The overall program goal was to reduce adolescent nicotine vaping and improve health outcomes by helping teens successfully quit e-cigarette use.

The primary evaluation question was whether participation in the intervention increased 30-day point-prevalence abstinence from vaping at the seven-month follow-up period. Secondary objectives included examining repeated abstinence over time and determining whether quitting vaping led adolescents to transition to combustible tobacco products.

Sampling Method

Identification of Sampling Method

The study used a non-probability convenience sampling method. Participants were recruited through advertisements placed on social media platforms including Instagram, Facebook, and Snapchat. Adolescents who viewed the advertisements voluntarily completed an online eligibility screening questionnaire. Eligible participants included U.S. residents aged 13 to 17 years who reported vaping within the past 30 days, expressed interest in quitting within 30 days, and owned a mobile phone with text messaging capabilities.

The sample included 1,503 adolescents randomized into intervention and assessment-only control groups. Participants represented diverse demographic backgrounds, including differences in race, gender identity, and sexual orientation. The average participant age was 16.4 years old, and many participants demonstrated moderate to high nicotine dependence.

Strengths and Limitations of the Sampling Method

One major strength of convenience sampling through social media was the researchers' ability to recruit a large national sample quickly and efficiently. Because adolescents frequently use social media, this recruitment method successfully reached the target population of teen e-cigarette users. The method was also cost-effective and appropriate for a digital intervention study.

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Another strength was that the sample included participants from diverse backgrounds. The study included substantial representation of sexual minority adolescents and adolescents with varying levels of nicotine dependence, which increased the relevance of findings for high-risk populations.

However, convenience sampling also created important limitations regarding representativeness. Because participants self-selected into the study, the sample may not accurately represent all adolescent e-cigarette users in the United States. Adolescents who volunteered for the study may have been more motivated to quit vaping than average teen users. Additionally, teens without social media access, reliable internet service, or interest in quitting were excluded from participation. These limitations reduce generalizability of the results.

Additional information that would have strengthened assessment of representativeness includes socioeconomic status, geographic distribution, parental education level, and urban versus rural residence. These characteristics could help determine whether the sample reflected the broader adolescent vaping population.

An alternative sampling approach that could have been used is stratified random sampling. Researchers could have recruited adolescents through schools across different geographic regions and demographic categories. Stratified random sampling would improve representativeness by ensuring proportional inclusion of important subgroups such as race, gender, socioeconomic status, and grade level.

The advantage of convenience sampling is that it is faster, less expensive, and easier to implement, especially for digital interventions. In contrast, stratified random sampling would provide stronger external validity and greater generalizability but would require significantly more resources, coordination, and time. Additionally, school-based recruitment may reduce anonymity and discourage honest disclosure of vaping behaviors among adolescents.

Evaluation Design

The study used a randomized clinical trial (RCT) design with parallel intervention and control groups. Participants were randomly assigned either to the vaping cessation text-message intervention or to an assessment-only control condition. The study also included a smaller waitlist control group to assess possible effects of repeated assessments and retention incentives.

Several components of the evaluation design strengthened the relationship between the intervention and the observed outcomes. First, random assignment reduced selection bias and increased internal validity by helping ensure baseline equivalence between groups. Table 1 demonstrated similar participant characteristics between groups at baseline.

Second, the double-blind design reduced bias because participants and data collection staff were unaware of treatment assignment. Third, the use of a control group allowed researchers to compare cessation outcomes between adolescents receiving the intervention and those receiving only assessment messages.

The study also used follow-up assessments at one month and seven months after randomization. These multiple follow-up periods allowed researchers to evaluate both short-term and longer-term cessation outcomes. The primary outcome measure was 30-day point-prevalence abstinence from vaping at seven months. Researchers analyzed outcomes using intention-to-treat analysis, meaning participants lost to follow-up were considered to still be vaping.

Overall, the randomized clinical trial design provided strong evidence that the text-message intervention contributed to improved vaping cessation outcomes among participants.

Data

The study primarily used quantitative data. Researchers collected numerical data through online surveys, telephone follow-ups, and text-message assessments. Quantitative methods allowed researchers to statistically compare vaping cessation outcomes between intervention and control groups.

Several constructs and measurement tools were used throughout the study. Researchers collected demographic data including age, gender, race, ethnicity, and sexual orientation. Vaping-related variables included vaping frequency, confidence to quit, motivation to quit, and previous quit attempts.

Nicotine dependence was measured using multiple validated scales, including:

- Penn State Electronic Cigarette Dependence Index (PSECDI)
- E-Cigarette Dependence Scale (EDS)
- E-cigarette Fagerström Test of Cigarette Dependence (e-FTCD)
- Hooked on Nicotine Checklist (HONC)

Researchers also measured psychosocial variables using the Global Appraisal of Individual Needs–Short Screener (GAIN-SS), Roberts UCLA Loneliness Scale (RULS), and Pediatric ACEs and Related Life Event Screener (PEARLS).

The primary outcome variable was self-reported 30-day abstinence from vaping at seven months. Secondary outcomes included repeated abstinence and use of combustible tobacco products. All outcome data were analyzed statistically using logistic regression and relative risk calculations.

Internal Validity

1. Attrition (Mortality)

One threat to internal validity was participant attrition. Some adolescents were lost to follow-up during the seven-month study period. If participants who dropped out differed significantly from those who remained, results could be biased.

The researchers mitigated this threat by using retention incentives and intention-to-treat analysis, which classified missing participants as continuing to vape. This conservative approach reduced the likelihood of overestimating intervention effectiveness.

2. Selection Bias

Selection bias is another threat because participants volunteered through social media advertisements. Adolescents motivated to quit vaping may have been more likely to enroll, potentially influencing results.w

Random assignment helped reduce this threat after enrollment by balancing participant characteristics between groups. However, selection bias may still limit how representative the sample was of all adolescent vapers.

3. Social Desirability Bias

Participants may have overstated quitting success because abstinence was self-reported rather than biochemically verified. Adolescents receiving supportive text messages may have felt pressure to report positive outcomes.

Researchers acknowledged this limitation and noted that biochemical verification is difficult in fully digital studies. Using anonymous surveys and blinded staff likely helped reduce some reporting bias.

4. History

The study occurred during the COVID-19 pandemic, which may have influenced vaping behaviors independently of the intervention. Increased public awareness of vaping risks, social isolation, or changing stress levels during the pandemic may have affected quitting rates.

Researchers partially addressed this issue by including a control group exposed to the same external events. Comparing both groups allowed researchers to better isolate intervention effects from broader societal influences.

External Validity

1. Limited Population Generalizability

One threat to external validity is that the study only included adolescents interested in quitting vaping. Results may not apply to adolescents who vape but have no intention of quitting.

To mitigate this issue, future studies could recruit both motivated and unmotivated adolescent vapers to determine whether the intervention works across varying readiness levels.

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2. Digital Access Bias

Because recruitment and intervention delivery relied on social media and mobile phones, adolescents without reliable internet access or texting capabilities were excluded. This may reduce generalizability to lower-income or technologically disconnected populations.

Researchers could improve external validity in future studies by combining digital recruitment with school-based or community-based recruitment methods to include adolescents with varying levels of technology access.

Stakeholder Involvement

The study demonstrated some stakeholder involvement through formative research with adolescents and young adults during development of the “This is Quitting” intervention. Messages included content written by users and tailored specifically to young people’s experiences and challenges with vaping cessation.

However, broader stakeholder involvement could have strengthened the evaluation study further. Important stakeholders could include parents, school administrators, pediatric healthcare providers, public health organizations, and community youth leaders. Including these stakeholders in planning and implementation could improve recruitment, increase program credibility, and support long-term sustainability.

For example, healthcare providers could promote the intervention during medical visits, while schools could assist with outreach and education. Parents could provide support systems for adolescents attempting to quit vaping. Increased stakeholder collaboration may also improve program dissemination and future implementation in real-world settings.

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Conclusion

Overall, the study used strong evaluation methods to examine the effectiveness of a text-message vaping cessation intervention for adolescents. The randomized clinical trial design, large sample size, validated measurement tools, and use of control groups strengthened the credibility of the findings. Results demonstrated that adolescents participating in the intervention were significantly more likely to quit vaping compared to those in the assessment-only control group.

Despite several limitations related to self-reporting, convenience sampling, and generalizability, the study provides important evidence supporting digital health interventions for adolescent vaping cessation. The intervention appeared effective across diverse demographic and psychosocial groups, including participants with high nicotine dependence.

If I were part of the evaluation research team, I would consider using a more representative sampling strategy, such as stratified random sampling through schools or healthcare settings, to improve generalizability. I would also explore adding biochemical verification methods when feasible and increasing stakeholder involvement from schools, healthcare providers, and families.

Overall, the study contributes meaningful evidence to adolescent tobacco prevention research and demonstrates the potential effectiveness of scalable digital interventions for public health behavior change.

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References

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